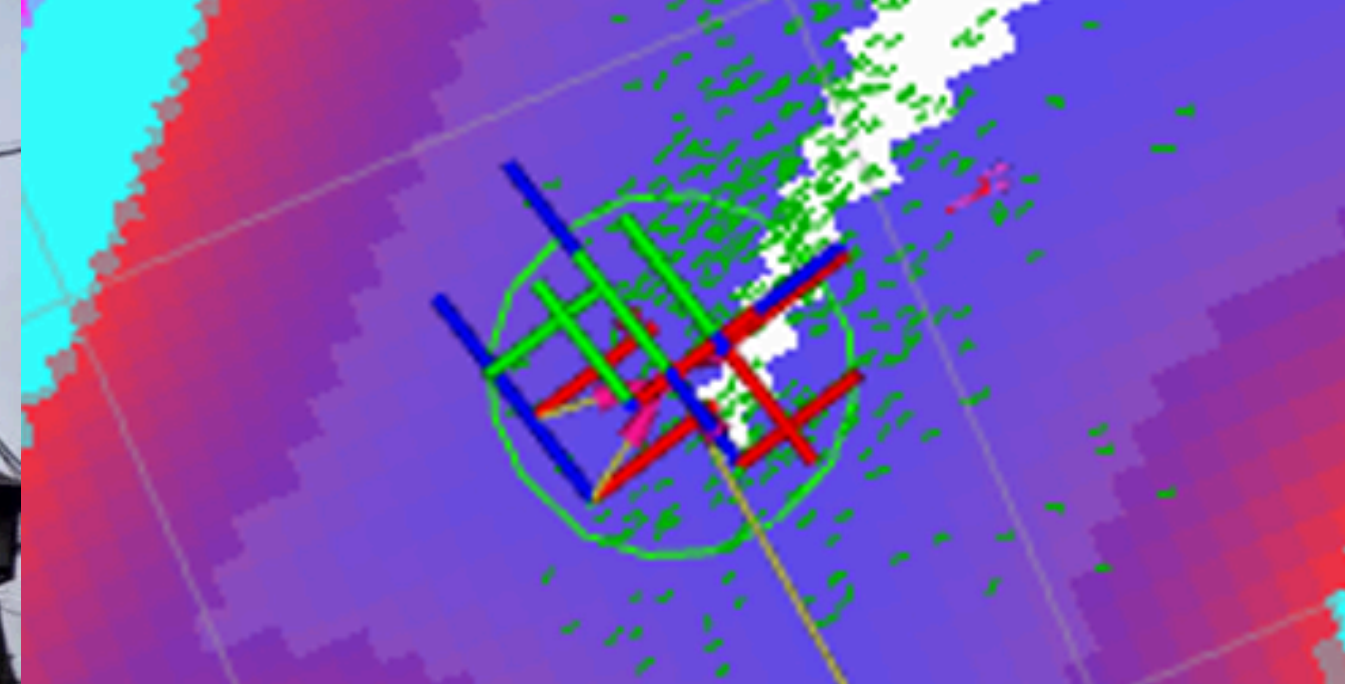




Autonomous Disaster Response Robot: Navigating Hazardous Terrains and Detecting Emergency Signals with TurtleBot3 Behavior Demos

Final Project Robotics Course
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I. INTRODUCTION

In disaster-affected areas, rubble and debris make

inputs and priorities. Together, these components allow the robot to navigate, detect emergency signals, and dynamically adjust its tasks, showcasing

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Keywords: TurtleBot3, ROS2, Behavior Trees, Disaster Robotics

What is our motivation and What is our project goals?

This project was inspired by the frequent occurrences of earthquakes in tectonically active regions, where destroyed homes and landscapes demand significant human effort and face challenges related to time and resource limitations.

The need for efficient and reliable solutions in such scenarios motivated the development of this autonomous robot.

By creating an autonomous robot capable of assisting in search operations, providing aid, or guiding individuals who may be lost, we hope to alleviate some of the burdens on human responders. This robot is inspired by the autonomous systems used by SAR (Search and Rescue) teams, aiming to replicate and enhance their capabilities in navigating challenging terrains and responding to emergencies.

With this project, we strive to make disaster response faster, safer, and more effective, ultimately saving lives and reducing human risk in critical situations.



ABC News - The Walt Disney Company
At least 8 dead after powerful earth...



ANTARA News Kalimantan Selatan
No report of Indonesians affected by ...



thesun.my
240 people still missing since New Yea...

What is our motivation and What is our project goals?

- Develop an autonomous TurtleBot3 robot for disaster response.
- Implement Behavior Trees for dynamic task management.
- Enable navigation in uneven terrains and detection of emergency signals
- Trying to learn to integrate RGB cameras, LiDAR, and ROS2 navigation for real-time adaptability.

Qlearning 1 : Perception

Description:

Perception, in this context, refers to the robot's ability to have a sense of the surrounding environment. Some people may call it mapping.

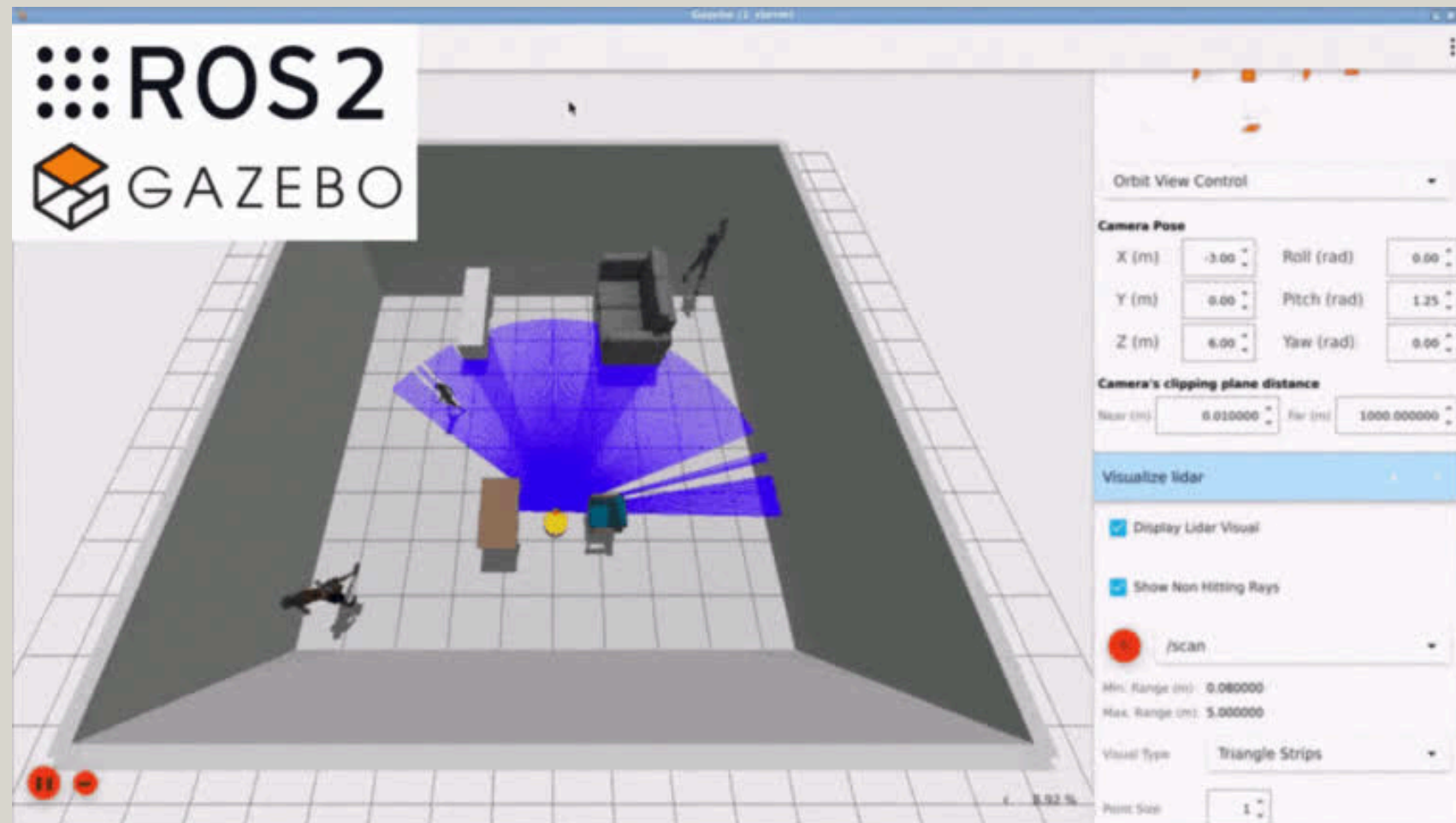
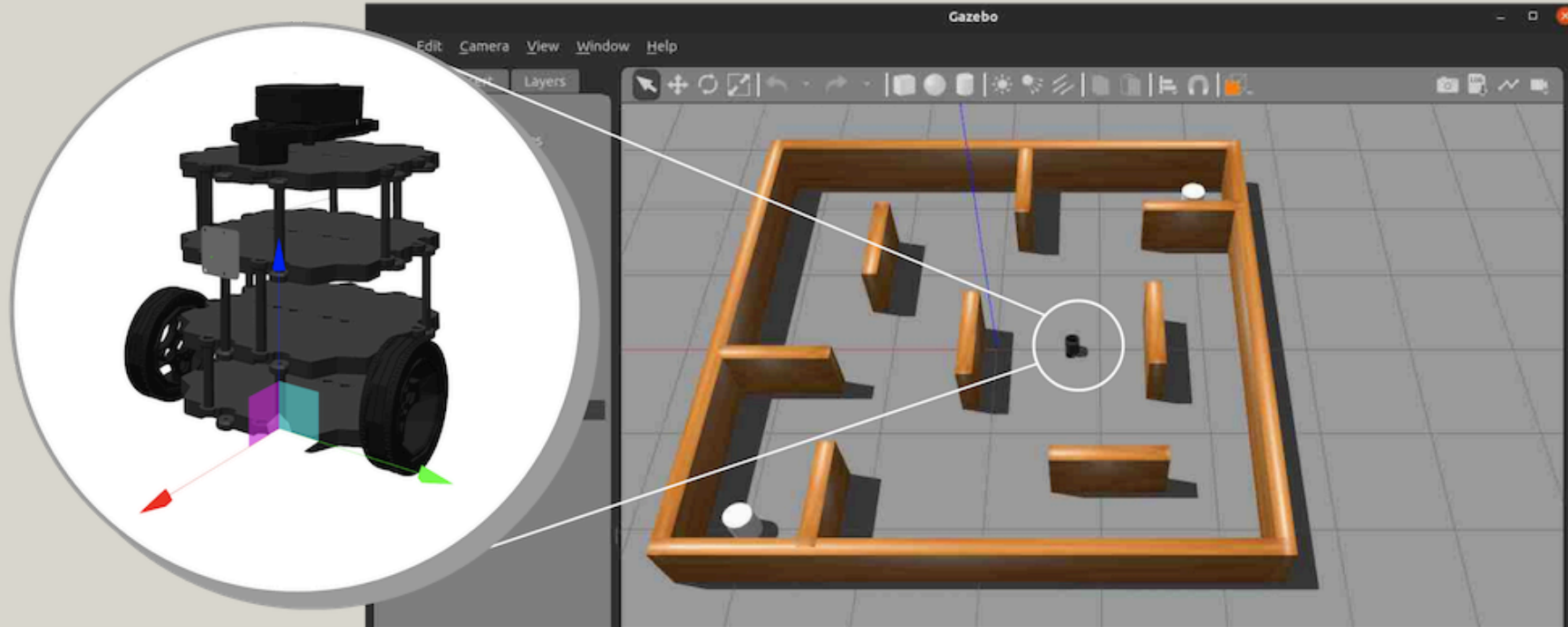
Key components:

Gazebo: for the simulation environment, also replicates disaster-like conditions.

Sensors: RGB camera, or LIDAR to provide environmental data for perception and decision-making

ROS2 Navigation Stack: For path planning and obstacle avoidance

Python's Behaviour Trees: To manage the robot's actions based on sensor inputs



Completion of Perception Component

Implemented systems:

Gazebo: To simulate and troubleshoot perception errors.

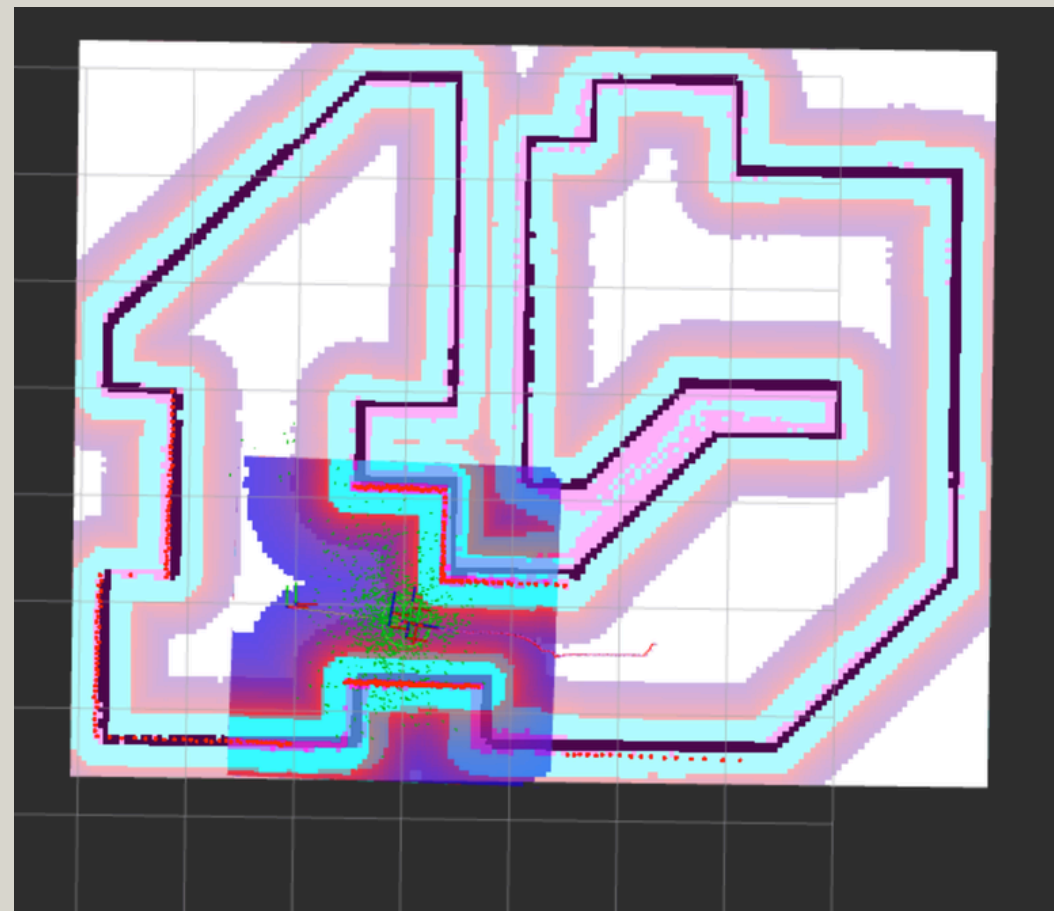
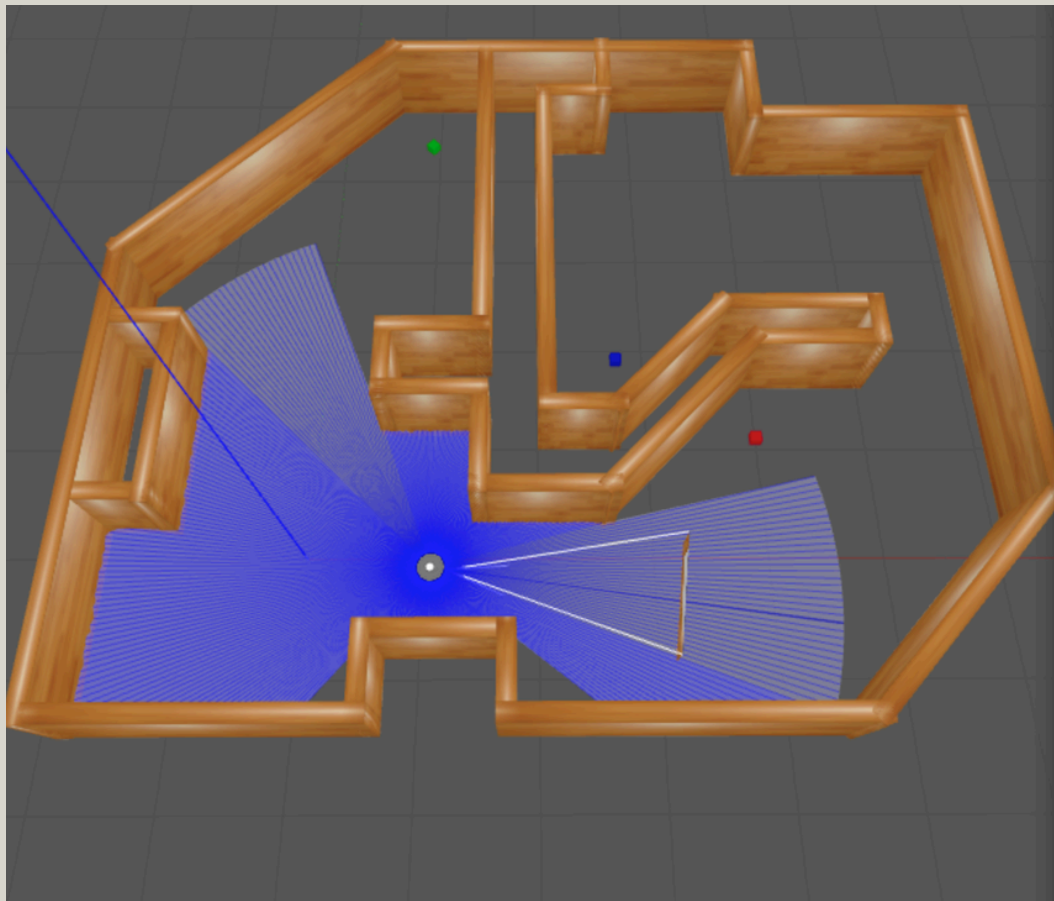
RGB Camera: for environment color detections and visual data

HSV color space threshold: As a threshold to filter colors and process visual data

LIDAR: provide detailed distance measurements to identify obstacles and avoid collision.

Achieved tasks:

- **Able to switch tasks**, such as navigations and obstacle avoidance based on sensor inputs.
- **Able to detect and identify emergency signals**, such as color markers, using its RGB camera.
- **Able to ensure safe and efficient movement**, particularly in dynamic and unpredictable environments like disaster zones
- **Able to interact effectively with its surroundings** and **perform tasks autonomously**.



- Gazebo Simulation

Qlearning 2 : Robot Navigation Component Overview

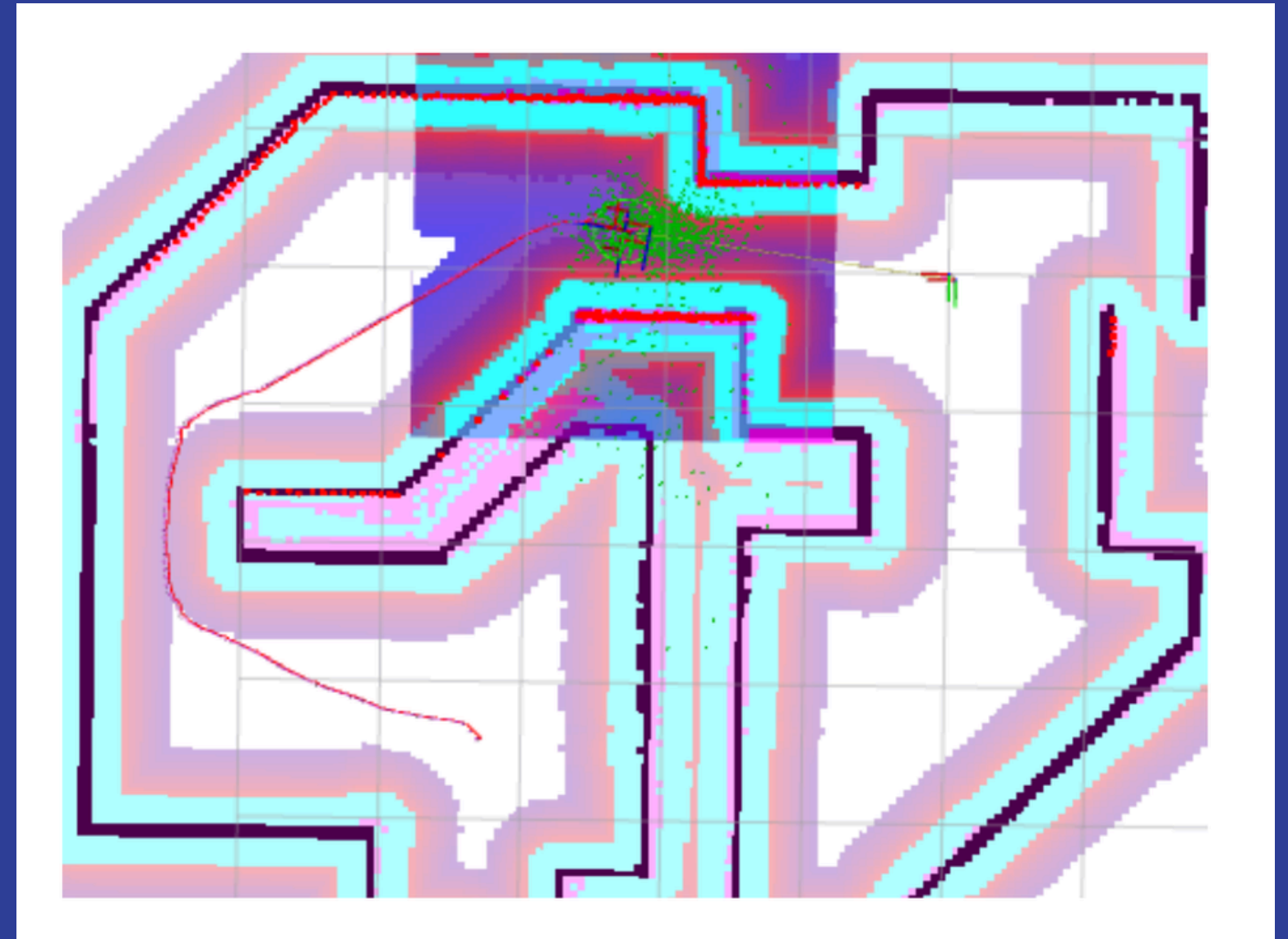
The navigation component allows the robot to autonomously traverse disaster terrains, avoiding obstacles and reaching targets efficiently. It adapts dynamically to unpredictable environments, ensuring reliable operation in search-and-rescue scenarios.

Core Components

- ROS2 Navigation Stack: Real-time path planning and obstacle avoidance.
- Behavior Trees (BTs): Enables dynamic task-switching and decision-making.
- Sensors:
 - LiDAR: Detects obstacles and maps distances.
 - RGB Camera: Identifies hazards and targets.

Key Metrics

- Path Completion Time: Time taken to reach a destination.
- Navigation Accuracy: Precision in reaching target locations.
- Distance Traveled: Efficiency of the chosen path.



"Simulated environment showcasing the robot's navigation path, planned routes, and obstacle detection using ROS2 and Gazebo."

Completion of Navigation Component

Objective

The goal was to develop a navigation system that allows the robot to move through hazardous terrains, avoid obstacles, and adapt to unpredictable environments.

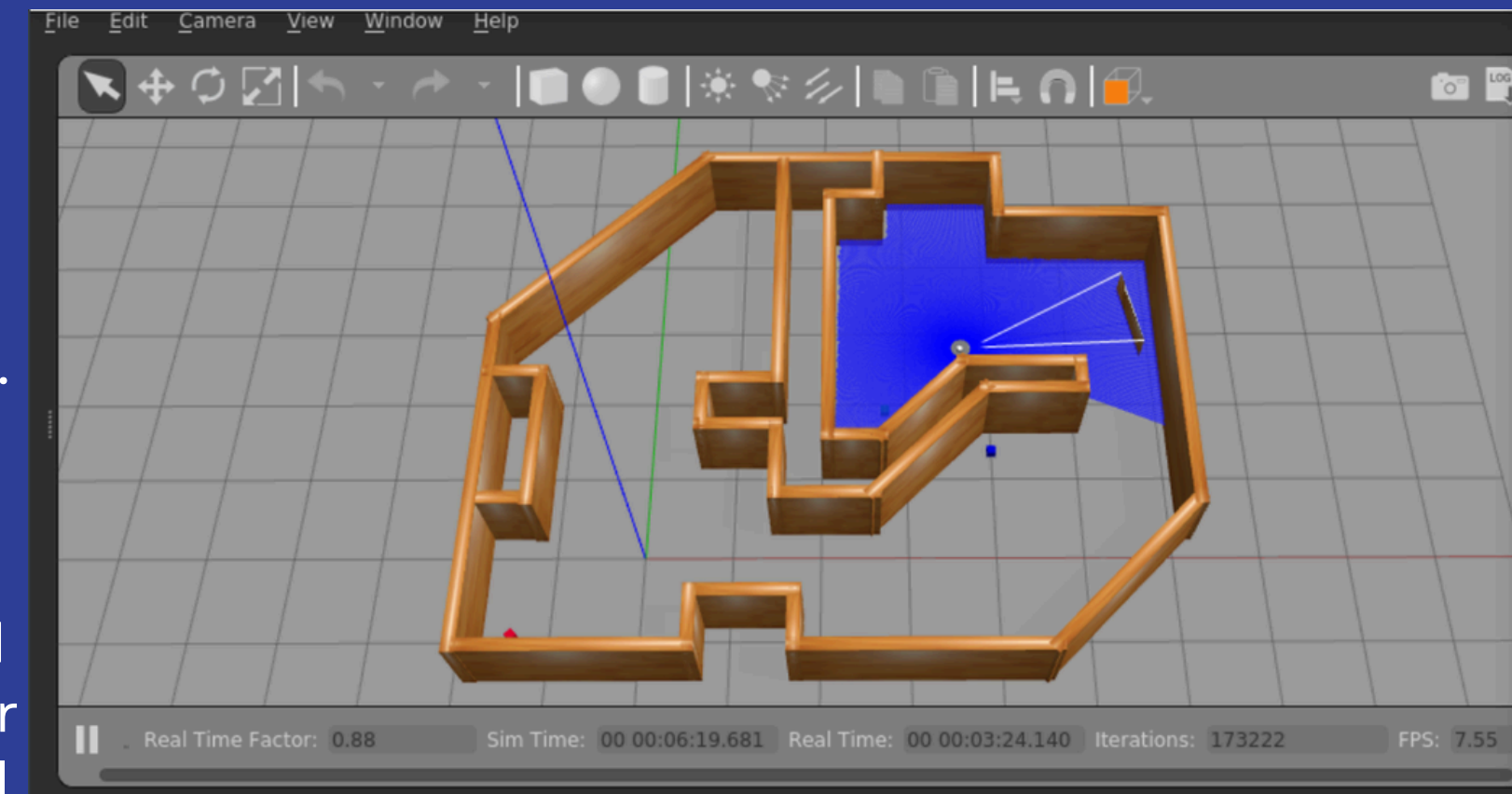
Key Achievements

- Successfully navigated through unmapped and dynamic environments.
- Avoided obstacles, both static and moving, with minimal collisions.
- Adapted well to disaster-like scenarios during testing.

The navigation system is built using Q-learning, where the robot decides its actions (like moving or stopping) based on its position and nearby obstacles. It uses tools like Gazebo for simulation and ROS2 for controlling movement. A reward system motivates it to reach the goal efficiently while avoiding penalties for collisions or delays.

Results

- The system proved effective in simulations, demonstrating smooth and reliable navigation.
- The robot consistently adapted to challenges, completing tasks efficiently.



“Simulated environment showcasing the robot’s navigation success in avoiding obstacles and completing routes safely.”

Qlearning 3 : Algorithm

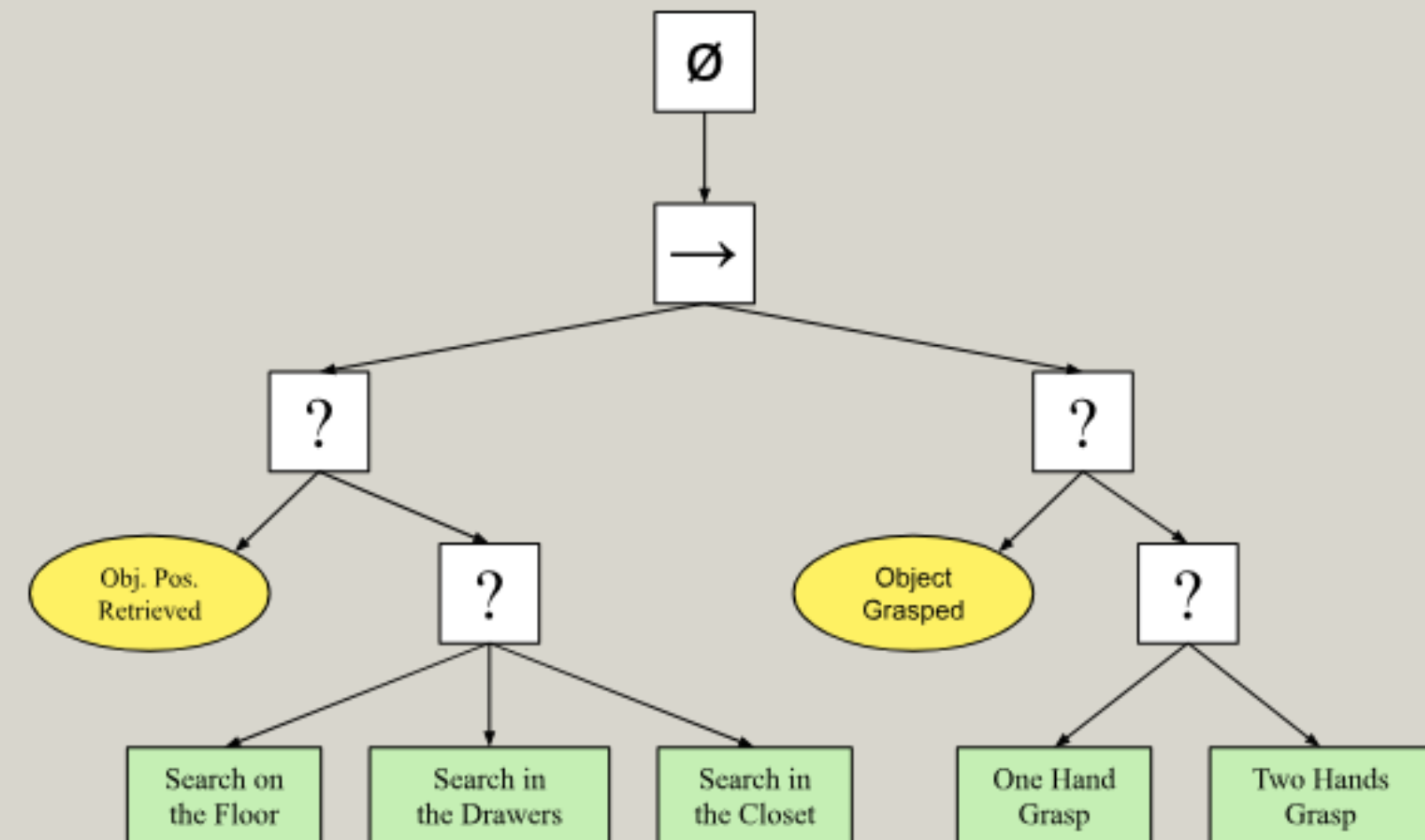
Utilizing Behavior Trees Algorithm for Autonomous Decision-Making

In this context of this project, our algorithm approach goes to BT. Behavior Trees (BTs) are used to control the TurtleBot3 for disaster response. BTs define the robot's behavior hierarchically, dividing it into modular tasks.

Behavior Trees offer several advantages in this case, including modularity, reactivity, and efficiency. Modularity allows tasks such as navigation, obstacle avoidance, and emergency signal detection to be managed independently, giving a clear and organized structure. Reactivity makes real-time task switching based on changes in the environment, enabling the system to adapt effectively.

Features:

- Implemented using the py_trees Python library.
- HSV color space thresholding for emergency signal detection.
- Dynamic task switching for smooth transitions between behaviors.
- Application: Essential for robots in disaster zones, where adaptability and efficiency are critical.



Completion of Behavior Trees Component

Open

autonomy_node.py

Save

```
15 import os
16 import yaml
17 import random
18 import rclpy
19 from rclpy.node import Node
20 import time
21 import py_trees
22 import py_trees_ros
23 from py_trees.common import OneShotPolicy
24 from ament_index_python.packages import get_package_share_directory
25
26 from tb3_behaviors.navigation import GoToPose, GetLocationFromQueue
27 from tb3_behaviors.vision import LookForObject
28
29
30 default_location_file = os.path.join(
31     get_package_share_directory("tb3_worlds"), "maps",
32     "sim_house_locations.yaml"
33 )
34
35 class AutonomyBehavior(Node):
36     def __init__(self):
37         super().__init__("autonomy_node")
38         self.declare_parameter("location_file",
39 value=default_location_file)
40         self.declare_parameter("tree_type", value="queue")
41         self.declare_parameter("enable_vision", value=True)
42         self.declare_parameter("target_color", value="blue")
43
44         # Parse locations YAML file and shuffle the location list.
45         location_file = self.get_parameter("location_file").value
46         with open(location_file, "r") as f:
47             self.locations = yaml.load(f, Loader=yaml.FullLoader)
48         self.loc_list = list(self.locations.keys())
49         random.shuffle(self.loc_list)
50
51         # Create and setup the behavior tree
52         self.tree_type = self.get_parameter("tree_type").value
53         self.enable_vision = self.get_parameter("enable_vision").value
54         self.target_color = self.get_parameter("target_color").value
55         self.create_behavior_tree(self.tree_type)
56
57         self.tree.node.get_logger().info(f"Using location file:
58 {location_file}")
```

Python Tab Width: 8 Ln 15, Col 10 INS

Environment Setup with Docker:

Docker Compose automates the setup of the simulation environment.

- docker compose up sim to launch the Gazebo simulation.
- docker compose up demo-behavior-py for Python-based behavior demos.

Core Scripts for Behavior Tree :

1. autonomy_node.py:

Main controller using Behavior Trees to manage robot actions hierarchically.

- Configures parameters (location_file, tree_type, target_color).
- Implements Naive Tree and Queue Tree for dynamic task execution.
- Functions like GoToPose for navigation and LookForObject for object detection.

2. test_move_base.py:

- Validates TurtleBot3’s navigation capability.
- Converts positional inputs (x, y, theta) into ROS2-compatible quaternion orientation.
- Tests navigation to predefined target locations independently.

3. test_vision.py:

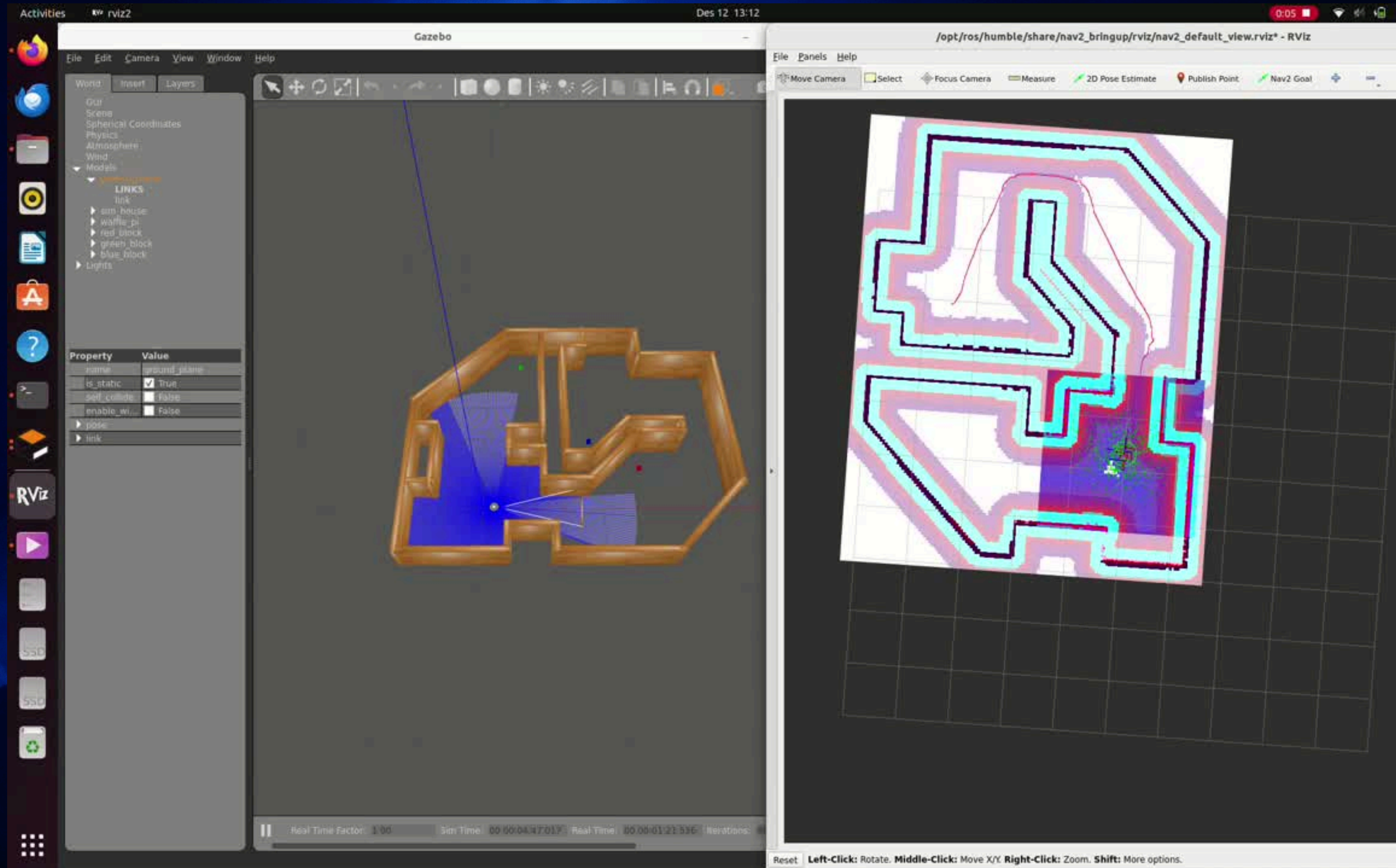
Detects emergency signals using HSV color thresholding.

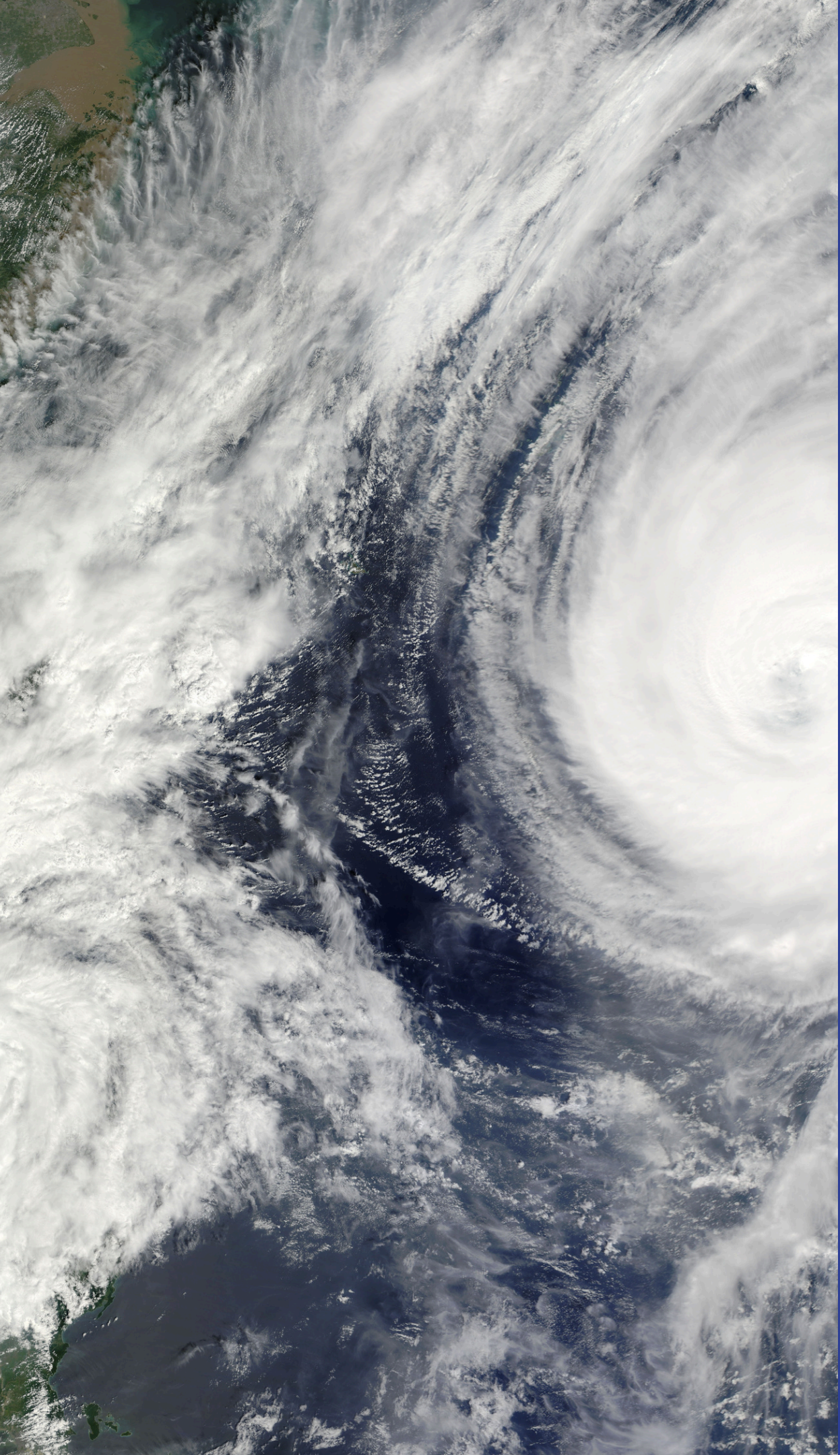
- HSV Thresholding for segmenting target color regions.
- Blob Detection to locate specific colored areas.
- ROS Subscription to access live camera data.
- Visualizes detection results via OpenCV to refine thresholds.

Script Integration:

These scripts, coupled with Docker’s reproducible setup, enable TurtleBot3 to autonomously navigate, detect colors, and adapt to dynamic tasks in disaster simulations.

Demo Video





Challenge

Our current robot simulation system faces several limitations. One major issue is its reliance on RGB cameras and LiDAR, which may struggle in real-world disaster scenarios where smoke, dust, or debris obstruct visibility.

Then, the system is designed for single-robot operation, that lacking coordination strategies for multi-robot setups, which presents challenges in real-time communication and dynamic task allocation among multiple robots.

The Gazebo simulation used for testing only provides basic disaster zone scenarios, failing to replicate the complexity of real-world conditions such as unstable terrain or extreme weather. Despite these limitations, the foundational concepts for autonomous disaster response robots have been established, though significant improvements are necessary.

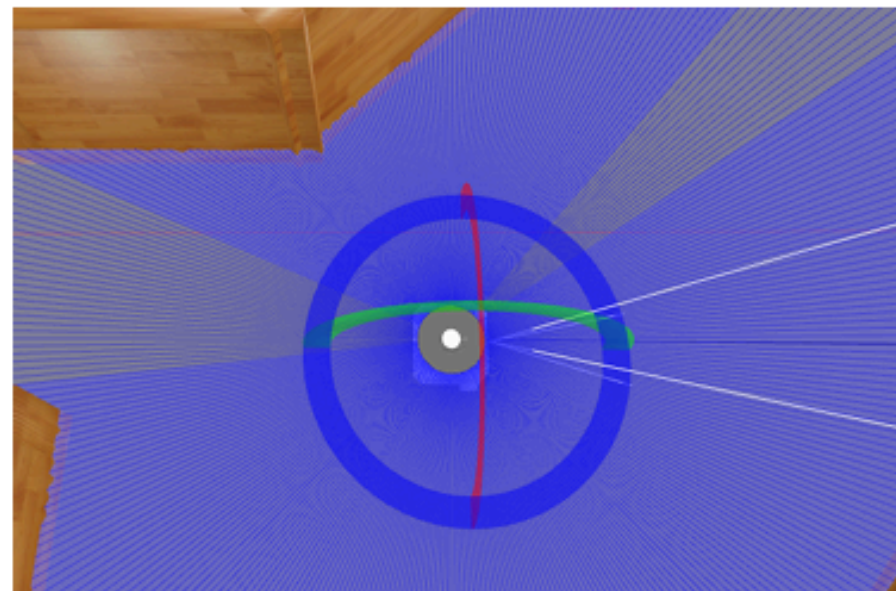
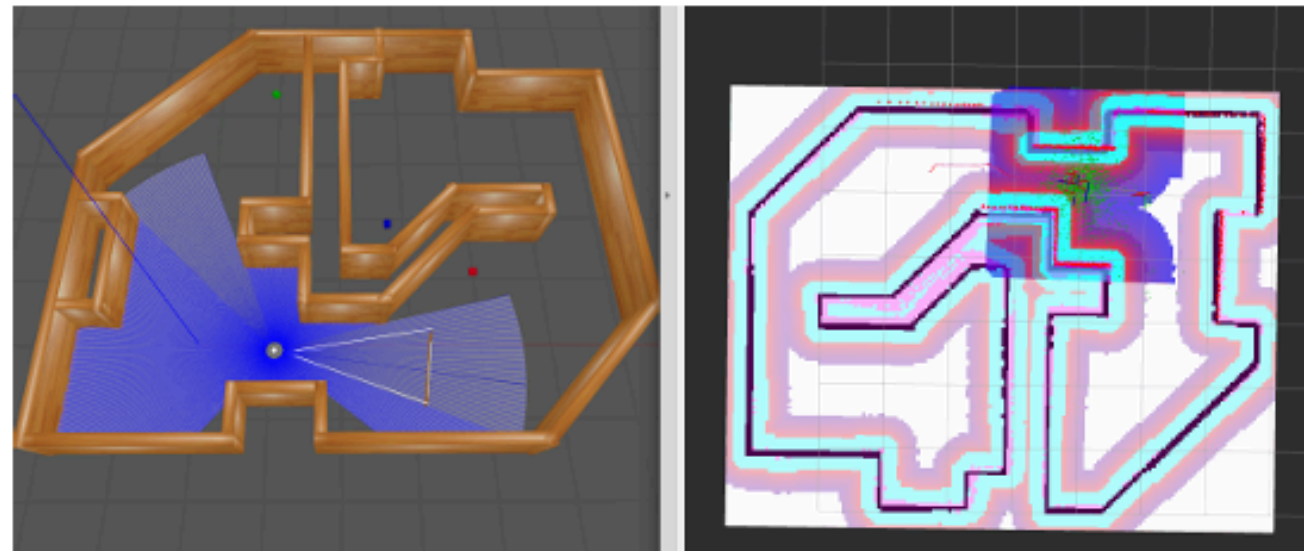
While the current Behavior Tree structures are functional, they need further refinement, and incorporating advanced algorithms like deep reinforcement learning could improve the robot's adaptability and decision-making capabilities.

Let's review.

What do we work?

We have successfully simulated the Autonomous Disaster Response Robot: Navigating Hazardous Terrains and Detecting Emergency Signals using the behavior tree algorithm on the TurtleBot3.

The simulation demonstrated the robot's ability to move and detect the terrain or map provided. Future work will focus on enhancing the map with real-world conditions and integrating additional features into the robot to improve navigation and detection capabilities in actual scenarios and learning more about



Picture 10
Gazebo and
Output Simulatio

What is our future work?

If we have several more weeks, the focus will be on improving the simulation in Gazebo to make it more realistic, including adding terrains like rubble and uneven surfaces. This will help the robot better adapt to disaster conditions. Alongside this, the team will research related robotics topics, such as behavior trees and sensor integration, while connecting with experts to gain more insights.

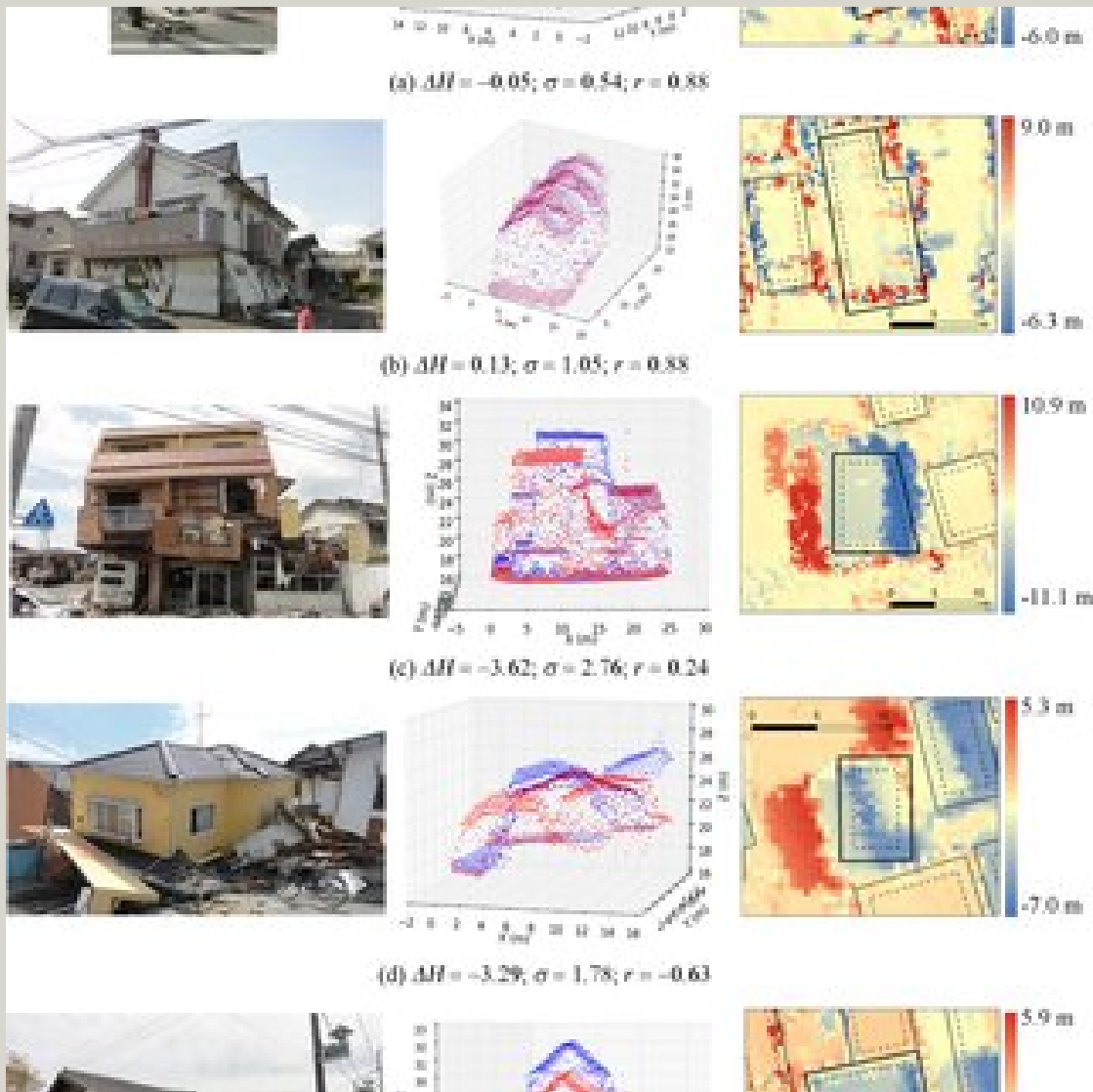
Although the current plan is ambitious, the team aims to align it with practical, existing technologies by focusing on achievable tasks, efforts will be directed at enhancing the simulation map. After that, the focus will shift to learning from academic papers and expert feedback to refine the project step by step.

Open Access Article

SAR and LIDAR Datasets for Building Damage Evaluation Based on Support Vector Machine and Random Forest Algorithms—A Case Study of Kumamoto Earthquake, Japan

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THANK YOU

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